

QUBE Motor Lab

1. Pre-Lab

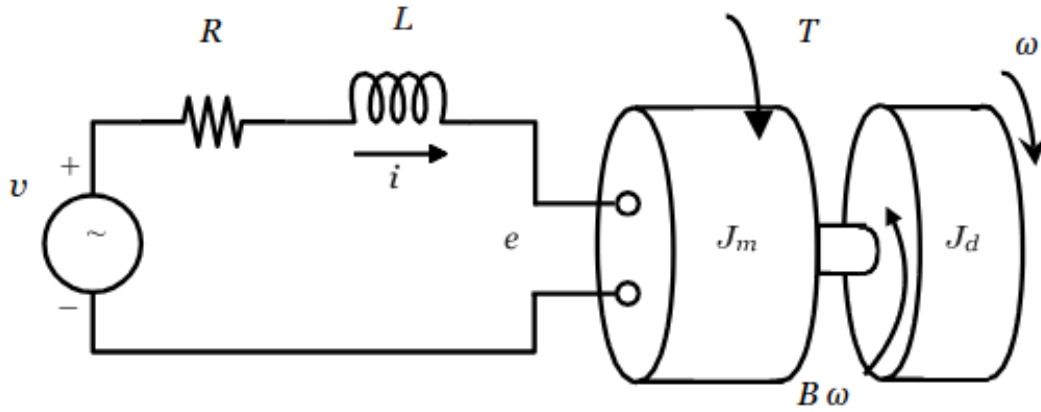


Figure 1: Schematic for the electrical and mechanical model of an electrical motor with disc (load) connected to the motor hub, adapted from “ME352-Qube-Lab1-Modeling-F25 1.pdf”.

1.1.

Note ($J = J_m + J_d$). Assume zero initial conditions. Let $Y(s)$ and $U(s)$ be the motor angular velocity and the applied voltage, respectively, in the Laplace domain.

The set of differential equations governing the system described in Figure 1 are:

$$L\dot{I} + RI + K_b\omega = v \quad (1)$$

$$J\dot{\omega} + B\omega = K_m I \quad (2)$$

Rearranging Equation 2 to isolate I on one side yields Equation 3:

$$K_m^{-1}(J\dot{\omega} + B\omega) = I \quad (3)$$

Substituting Equation 3 into Equation 1 yields Equation 4:

$$\frac{L}{K_m}(J\ddot{\omega} + B\dot{\omega}) + \frac{R}{K_m}(J\dot{\omega} + B\omega) + K_b\omega = v \quad (4)$$

This can be rewritten as:

$$\left(\frac{LJ}{K_m}\right)\ddot{\omega} + \left(\frac{BL + JR}{K_m}\right)\dot{\omega} + \left(\frac{BR + K_b K_m}{K_m}\right)\omega = v$$

Taking the Laplace Transform of Equation 4 yields:

$$\left(\frac{LJ}{K_m}\right)s^2 Y(s) + \left(\frac{LB + RJ}{K_m}\right)s Y(s) + \left(\frac{RB + K_b K_m}{K_m}\right)Y(s) = U(s) \quad (5)$$

Note Equation 5 be rewritten as:

$$Y(s) \left(\frac{(LJs^2) + (LB + RJ)s + RB + K_b K_m}{K_m} \right) = U(s)$$

Then, by dividing both sides of Equation 5 by $Y(s)$, and then taking the reciprocal of both sides, we get the transfer function:

$$\frac{Y(s)}{U(s)} = \frac{K_m}{LJs^2 + (LB + RJ)s + RB + K_b K_m} \quad (6)$$

1.2.

Now, consider the state vector $\left(x = \begin{bmatrix} \omega \\ I \end{bmatrix} \right)$, with output ω . Then, $\left(\dot{x} = \begin{bmatrix} \dot{\omega} \\ \dot{I} \end{bmatrix} \right)$. By rearranging Equation 2, we can express $\dot{\omega}$ as a linear combination of elements in our state vector:

$$\dot{\omega} = -\left(\frac{B}{J}\right)\omega + \left(\frac{K_m}{J}\right)I \quad (7)$$

This can also be achieved for $\dot{I}$ by rearranging Equation 1:

$$\dot{I} = \left(-\frac{K_b}{L}\right)\omega + \left(-\frac{R}{L}\right)I + \frac{v}{L} \quad (8)$$

Therefore, we can express this system with the following state-space matrix equations:

$$\dot{x} = \begin{bmatrix} \dot{\omega} \\ \dot{I} \end{bmatrix} = \begin{bmatrix} -\frac{B}{J} & \frac{K_m}{J} \\ -\frac{K_b}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} \omega \\ I \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} v \quad (9)$$

$$y = [\omega] = [1 \ 0] \begin{bmatrix} \omega \\ I \end{bmatrix} + [0]v \quad (10)$$

1.3.

Now, consider the system where $((L = 0) \wedge (B = 0))$. Then, Equation 1 and Equation 2 become:

$$RI + K_b \omega = v \quad (11)$$

$$J\dot{\omega} = K_m I \quad (12)$$

Rearranging Equation 12 to isolate I on one side yields:

$$I = \frac{J\dot{\omega}}{K_m} \quad (13)$$

Substituting Equation 13 into Equation 11 yields:

$$\left(\frac{RJ}{K_m}\right)\dot{\omega} + K_b \omega = v \quad (14)$$

Taking the Laplace Transform of Equation 14 yields:

$$\left(\frac{RJ}{K_m}\right)sY(s) + K_b Y(s) = Y(s) \left(\frac{RJ s + K_b K_m}{K_m}\right) = U(s) \quad (15)$$

Then, by dividing both sides of Equation 15 by $Y(s)$, and then taking the reciprocal of both sides, we get the transfer function:

$$\frac{Y(s)}{U(s)} = \frac{K_m}{RJ s + K_b K_m} \quad (16)$$

1.4.

Now consider the state vector ($x = [\omega]$) with output ω . Then ($\dot{x} = [\dot{\omega}]$). By rearranging Equation 12 to isolate $\dot{\omega}$, we get:

$$\dot{\omega} = \left(\frac{K_m}{J} \right) I \quad (17)$$

However, note that, by rearranging Equation 11, we can express I as:

$$I = \left(-\frac{K_b}{R} \right) \omega + \left(\frac{v}{R} \right) \quad (18)$$

Substituting Equation 18 into Equation 17 yields:

$$\dot{\omega} = \left(-\frac{K_b K_m}{RJ} \right) \omega + \left(\frac{K_m v}{RJ} \right) v \quad (19)$$

Therefore, we can express this system with the following state-space matrix equations:

$$\dot{x} = [\dot{\omega}] = \left[-\frac{K_b K_m}{RJ} \right] [\omega] + \left[\frac{K_m v}{RJ} \right] v \quad (20)$$

$$y = \omega = [1][\omega] + [0]v \quad (21)$$

1.5.

Assume ($B = 0$) From the problem statement, we have the following givens:

$$R = 8.4\Omega, \quad L = 0.00116H, \quad J_m = (4.65 * 10^{-6}) \frac{\text{kg}}{m^2}, \quad K_m = 0.042, \quad K_b = 0.042$$

$$m = 0.053 \text{ kg}, \quad r = 0.0248m$$

Since J_d is modeled as a disc of mass m and radius r :

$$J_d = \frac{1}{2} m r^2 = (1.630 * 10^{-5}) \frac{\text{kg}}{m^2}$$

Recall that ($J = J_d + J_m$). Consequently,

$$J = J_d + J_m = (2.095 * 10^{-5}) \frac{\text{kg}}{m^2}$$

These numbers are quite decrepid. We can use Python (see Appendix Code in Section 3) to let the computer calculate the poles s_1 and s_2 for us:

$$s_1 = -7231.341 \text{ s}, \quad s_2 = -10.038 \text{ s} \quad (22)$$

Note that ($s_1, s_2 \in \mathbb{R}_{<0}$) implies that the system is stable. Additionally, since ($|s_1| < |s_2|$), s_1 is the dominant pole of the system, and consequently governs the system's response.

1.6.

Recall that τ is the time constant of a first-order system such that the transfer function $G(s)$ takes the form:

$$G(s) = \frac{K}{\tau s + 1}$$

where K is the DC gain of the system.

Consider Equation 16:

$$\frac{Y(s)}{U(s)} = \frac{K_m}{RJ s + K_b K_m}$$

This equation can be rewritten by dividing both the numerator and denominator by $K_b K_m$, yielding:

$$\frac{Y(s)}{U(s)} = \frac{\left(\frac{1}{K_b}\right)}{\left(\frac{RJ}{K_b K_m}\right) + 1} \quad (23)$$

From Equation 23, we can determine that:

$$K = \frac{1}{K_b} = (23.810) \frac{\text{rad}}{\text{s} \cdot \text{V}}, \quad \tau = \frac{RJ}{K_b K_m} = (9.976 * 10^{-2}) \text{ s} \quad (24)$$

Therefore, it can be concluded that the motor reaches 63.2% of its steady state angular speed in about 0.10s, and for every 1V of input, the maximum steady state angular speed increases by $(23.810) \frac{\text{rad}}{\text{s} \cdot \text{V}}$.

2. Lab

My lab partner was Rudis Jatnieks.

2.1.

Let $\omega_{\text{actual}, xV}$ be the rotational velocity of the QUBE motor with an input square wave voltage of xV . Using the values for $\omega_{\text{actual}, 12V}$ derived from Figure 6, we found that our time constant and DC gain, respectively, were:

$$\tau_{\text{actual}} = 156.024 \text{ ms} = 1.560 * 10^{-1} \text{ s}, \quad K = \left(\frac{321.3}{12} \right) \left(\frac{\text{rad}}{\text{s}} \right) V^{-1} = (26.775) \frac{\text{rad}}{\text{s} \cdot V} \quad (25)$$

2.2.

See the following page for the collected figures from our test runs at input voltages 0.5V, 6V, 12V, and 25V.

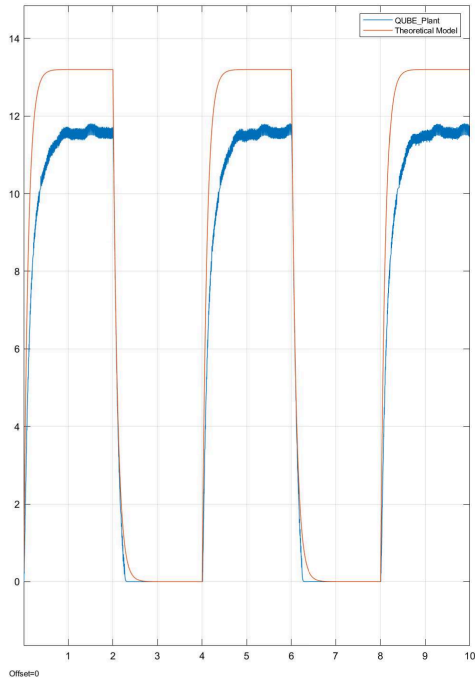


Figure 2: *Theoretical and Actual Rotational Velocity (in rad/s) vs. Time (in s) for 0.5V Input Voltage*

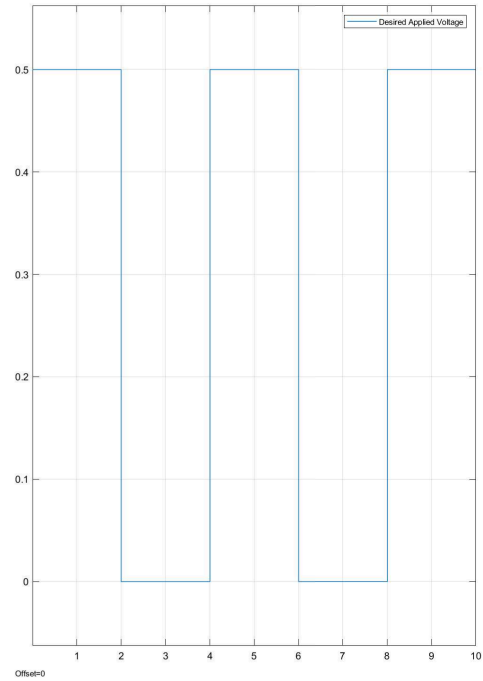


Figure 3: *0.5V Input Square Wave Voltage vs. Time (in s)*

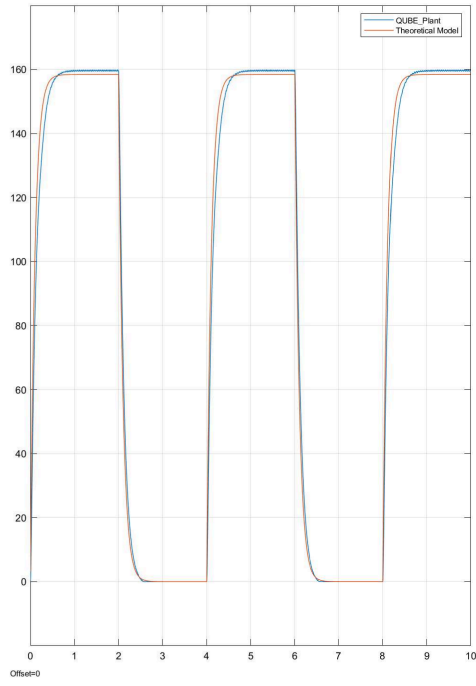


Figure 4: *Theoretical and Actual Rotational Velocity (in rad/s) vs. Time (in s) for 6V Input Voltage*

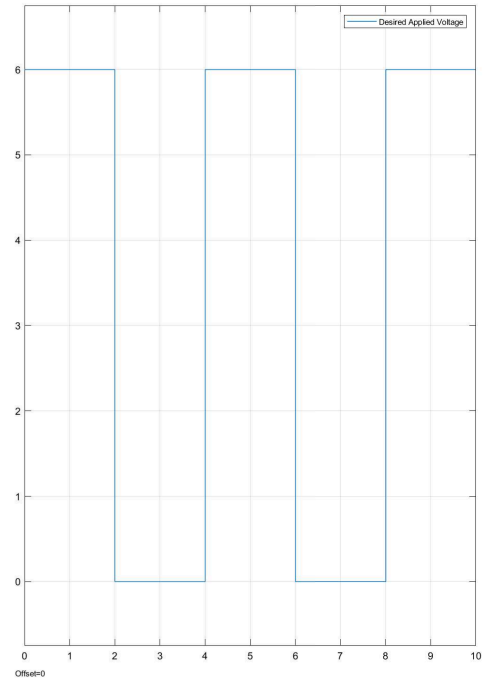


Figure 5: *6V Input Square Wave Voltage vs. Time (in s)*

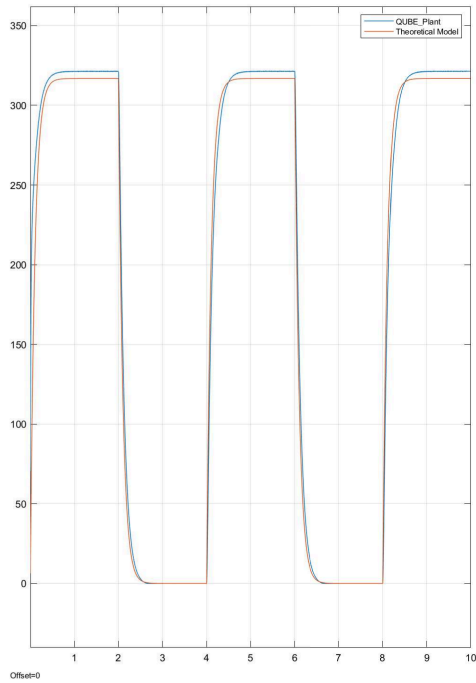


Figure 6: *Theoretical and Actual Rotational Velocity (in rad/s) vs. Time (in s) for 12V Input Voltage*

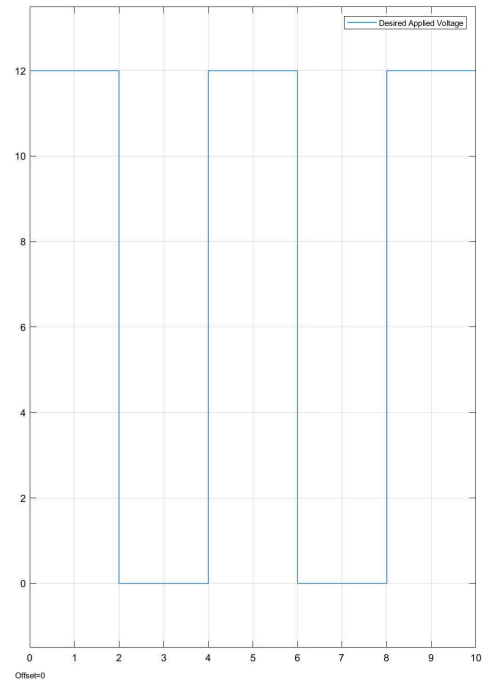


Figure 7: *12V Input Square Wave Voltage vs. Time (in s)*

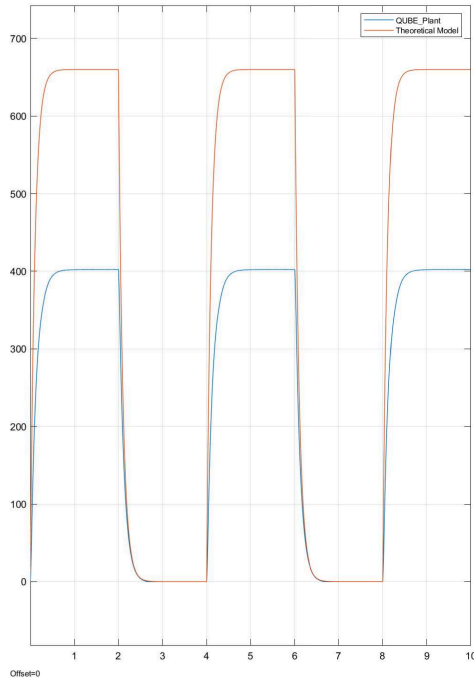


Figure 8: *Theoretical and Actual Rotational Velocity (in rad/s) vs. Time (in s) for 25V Input Voltage*

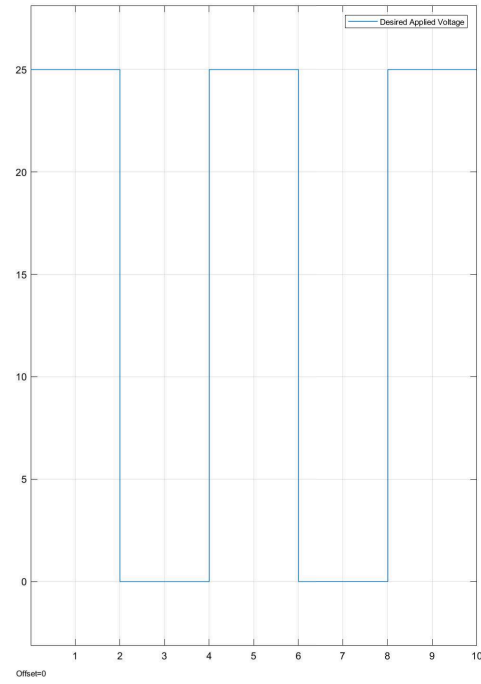


Figure 9: *25V Input Square Wave Voltage vs. Time (in s)*

At input voltages within the QUBE motor's rated operating range, the actual rotational speed of the motor follows the theoretical model. This is apparent in Figure 4 and Figure 6, where the theoretical and actual rotational velocities nearly overlap for most of the motion, indicating that the model predicts the motor behavior well under normal operating conditions. However, a small difference is still visible during the step response: when the voltage is applied, the actual motor response rises slightly differently from the theoretical curve and settles to a slightly different steady-state speed. By contrast, during the free response, when the input returns to 0V, the actual and theoretical velocities decay back to zero in a much more similar manner, with only minor deviation between the two curves.

2.3.

At low input voltages, the QUBE motor rotates at a speed lower than its theoretical speed, though the overall shape of the experimental response follows the theoretical model closely. As shown in Figure 2, the actual rotational velocity rises and falls in the same general pattern as the theoretical prediction, but its steady-state value remains consistently below the model during each high-voltage interval.

At high input voltages, the QUBE motor will rotate at a speed much lower than its theoretical speed, as overvolting the motor only increases its rotational speed slightly from its theoretical speed at its rated voltage (the QUBE motor is rated for 12V). This is apparent in Figure 8, as there is a significant space between the theoretical rotational speed of the QUBE motor and the actual rotational speed of the QUBE motor.

2.4.

Recall that the theoretical time constant and DC gain of the QUBE motor (from Equation 24) were:

$$K = \frac{1}{K_b} = (23.810) \frac{\text{rad}}{\text{s} \cdot \text{V}}, \quad \tau = \frac{RJ}{K_b K_m} = (9.976 * 10^{-2}) \text{ s}$$

Using the values from Equation 25, we can calculate the percent difference between the QUBE motor's actual and theoretical values for time constant and DC gain, respectively:

$$100\% \left(\frac{|1.560 * 10^{-1} \text{ s} - (9.976 * 10^{-2}) \text{ s}|}{9.976 * 10^{-2} \text{ s}} \right) \approx 56.396\% \quad (26)$$

$$100\% \left(\frac{|(26.775) \frac{\text{rad}}{\text{s} \cdot \text{V}} - (23.810) \frac{\text{rad}}{\text{s} \cdot \text{V}}|}{(23.810) \frac{\text{rad}}{\text{s} \cdot \text{V}}} \right) = 12.455\% \quad (27)$$

Within the QUBE motor's rated operating range, the actual and theoretical values of the time constant and DC gain were expected to be approximately the same. In reality, the motor's time constant and gain differed from their theoretical values by approximately 56% and 12%, respectively. Note that these actual values were derived from using the trace tool within the Simulink platform, which has limited precision. Instead, using a Matlab script to find the QUBE motor's actual DC gain K (by finding the maximum rotational velocity of the QUBE motor in the timeframe in which it was ran, and dividing that figure by the input voltage v of the system), and using that figure to find the time t where the motor first reaches a rotational speed of $((0.632Kv) \frac{\text{rad}}{\text{s}})$ would be more accurate for future experiments.

3. Appendix Code:

The following code was used to determine the poles of the system, and optionally graph them on $\mathbb{C}^2$.

```
import numpy as np
import matplotlib.pyplot as plt

# Motor parameters from the pre-lab
R = 8.4          # ohm
L = 0.00116      # H
Jm = 4.65e-6     # kg*m^2
Km = 0.042       # N*m/A
Kb = 0.042       # V/(rad/s)
B = 0.0          # viscous damping neglected for part (e)

# Disc parameters
m_disc = 0.053   # kg
r_disc = 0.0248  # m

# Disc inertia: Jd = (1/2) m r^2
Jd = 0.5 * m_disc * r_disc**2

# Total inertia
J = Jm + Jd

# Second-order transfer function denominator:
# G(s) = Km / [LJ s^2 + (LB + RJ) s + (RB + Kb*Km)]
den = [L * J, L * B + R * J, R * B + Kb * Km]

# Find poles
poles = np.roots(den)

# Print results
print(f"Disc inertia Jd = {Jd:.8e} kg*m^2")
print(f"Total inertia J = {J:.8e} kg*m^2")
print("\nDenominator coefficients:")
print(f"a2 = {den[0]:.8e}")
print(f"a1 = {den[1]:.8e}")
print(f"a0 = {den[2]:.8e}")

print("\nPoles of the system:")
for i, p in enumerate(poles, start=1):
    print(f"pole {i} = {p:.8f}")

# Optional: identify dominant pole
dominant_pole = poles[np.argmax(np.real(poles))] # closest to the imaginary axis
print(f"\nDominant pole = {dominant_pole:.8f}")

# Optional: plot poles on the complex plane
plt.figure(figsize=(6, 6))
plt.axhline(0, linewidth=1)
plt.axvline(0, linewidth=1)
plt.plot(np.real(poles), np.imag(poles), 'x', markersize=10, label='Poles')
plt.xlabel('Real')
plt.ylabel('Imaginary')
plt.title('Pole Locations of DC Motor System')
```

```
plt.grid(True)
plt.legend()
plt.show()
```